VIVEKANAND EDUCATION SOCIETY

INSTITUTE OF TECHNOLOGY (AUTONOMOUS)

Department of Electronics and Computer Science

in Association with Tinkerers Lab of ECS & VESIT IQAC and VESIT IIC.

TINKERER'S LAB ECS

NEXUS

CONNECTING MINDS TO CREATE SOLUTIONS

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Details

- TRACK: SENIOR TRACK
- DOMAIN: AIML
- PROBLEM STATEMENT: WAREHOUSE PACKING MANAGEMENT SYSTEM (PS2)
- TEAM NAME: DECODERS
- TEAM MEMBERS: RUJUL SALUNKHE, ATHARVA PHADTARE, HARSHITA ANCHAN, AVANTIKA SHARMA

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Problem Statement-WAREHOUSE PACKING MANAGEMENT SYSTEM

The Problem: No Digital Traceability

No Video Evidence

Packing sessions lack recorded video, making disputes impossible to resolve and creating accountability gaps.

Paper-Based Records

Paper-Based Records
Manual reports on paper create unstructured data with no searchable history or digital verification.

No Unified System

No Unified System
Operator, batch, count, and timing data exist in isolation — impossible to tie together for verification.

Why this matters:

1. Warehouse operations require traceability, every packing session needs to be accountable
2. Without a system that records and integrates all this data, errors compound and go unresolved
3. A packing record that cannot be verified is not useful as proof for clients, auditors, or managers

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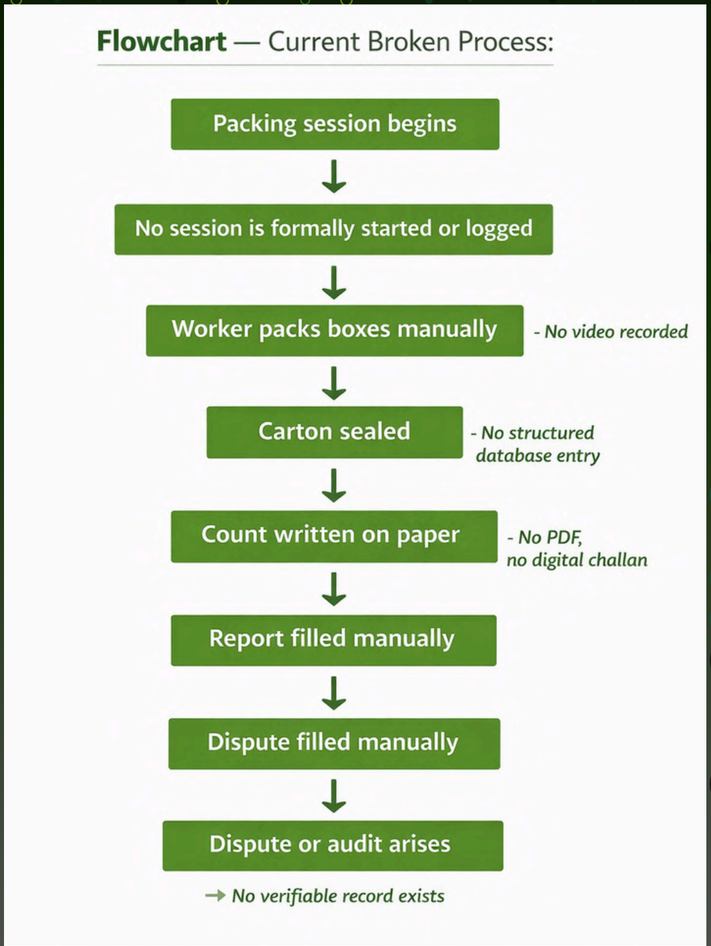
Problem Statement-WAREHOUSE PACKING MANAGEMENT SYSTEM

Current Reality

1. No formal session start or stop
2. No video recording
3. Paper-based count records
4. No packing report
5. No system integration

What is Needed

1. Managed packing sessions with timestamps
2. Compressed video stored per session on Linux
3. Structured database: Operator ID, Batch ID, count, logs
4. Auto-generated PDF challan on session stop
5. Web app connected to Linux backend in real time



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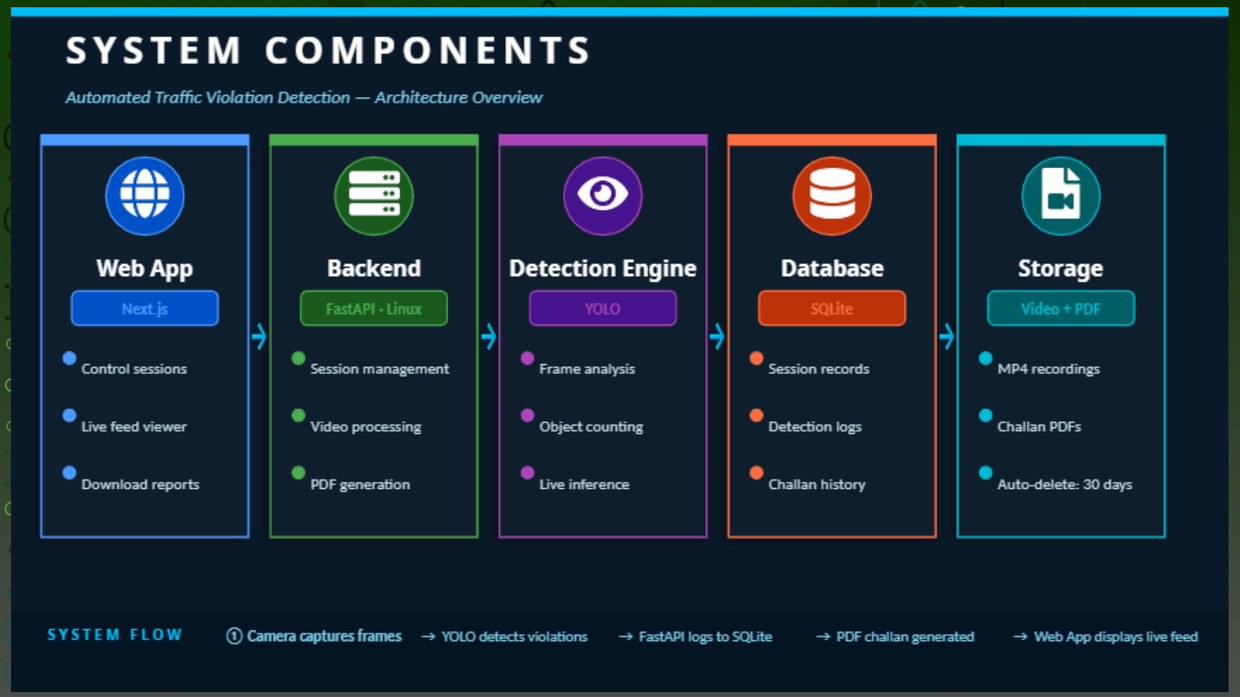
Proposed Idea/ Solution

Full Packing Session Management System

Web app controls sessions, Linux backend handles detection, storage, and auto-generates reports — fully offline.



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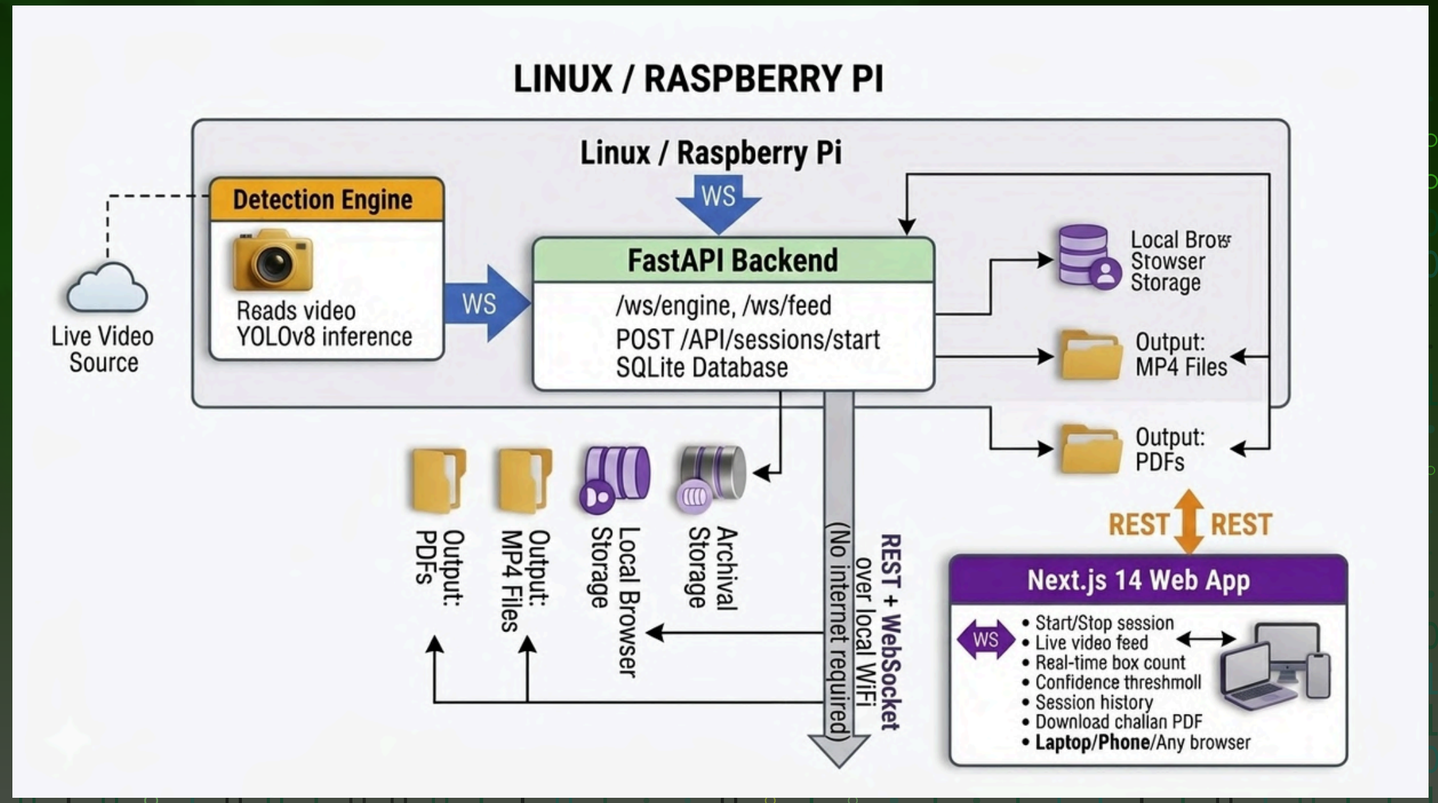
Technical Approach



The web app never communicates with the detection engine directly. All traffic, frames, counts, session control, file downloads goes exclusively through the FastAPI backend running on Linux.

Technologies used:

- **Languages**: JavaScript, Python.
- **Frontend**: Next.js, Shadcn
- **Backend**: Node.js, Web3.js.
- **Database**: SQLite

Hardware used:

- Board: Raspberry Pi 3B+
- Camera: Raspberry Pi Camera

System Architecture — Web App Communicates with Linux Backend

Technology stack

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Impacts and Benefits

IMPACTS

Session Traceability
Start/stop sessions with full tracking

Real-Time Visibility
Live video & count on any device

Instant Documentation
Auto PDF challan, no paperwork

Permanent Logging
Operator, batch, time, count stored

Verifiable Evidence
30-day video proof with overlay

BENEFITS

Low Cost System
~₹7,000 vs expensive industrial setups

No Cloud Dependency
Works offline, zero subscriptions

All-in-One Solution
Tracking, recording, logging, reports

Searchable History
Access sessions from any browser

Audit Ready Records
PDF challan for disputes & audits

UN SDGs

8 DECENT WORK AND ECONOMIC GROWTH

Decent Work & Economic Growth
Improves workforce efficiency and transparency

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE

Industry, Innovation & Infrastructure
Affordable, scalable industrial automation solution

12 NACHHALTIGE/R KONSUM UND PRODUKTION

Responsible Consumption & Production
Reduces errors, fraud, and packaging waste

Research and References

[1] I. F. E. Babila, S. A. E. Villasor and J. C. Dela Cruz, "Object Detection for Inventory Stock Counting Using YOLOv5," 2022 IEEE 18th International Colloquium on Signal Processing & Applications (CSPA), Selangor, Malaysia, 2022, pp. 304-309, doi: 10.1109/CSPA55076.2022.9782028.

[2] U. Obu, Y. Ambekar, H. Dhote, S. Wadbudhe, S. Khandelwal and S. Dongre, "Crop Disease Detection using Yolo V5 on Raspberry Pi," 2023 3rd International Conference on Pervasive Computing and Social Networking (ICPCSN), Salem, India, 2023, pp. 528-533, doi: 10.1109/ICPCSN58827.2023.00092.

[3] F. López, F. J. Torres, V. A. Ramírez, D. A. Núñez, R. Corona and A. R. López, "Raspberry Pi for Implementation of Web Technology in an Automation Process," 2019 IEEE International Autumn Meeting on Power, Electronics and Computing (ROPEC), Ixtapa, Mexico, 2019, pp. 1-6, doi: 10.1109/ROPEC48299.2019.9057040.

[4] Yesbox, "Box Detection Dataset," Roboflow Universe, 2026. [Online]. Available: https://universe.roboflow.com/yesbox/box_detection-69a4m-vycta-trcvg-flvdv/dataset/1

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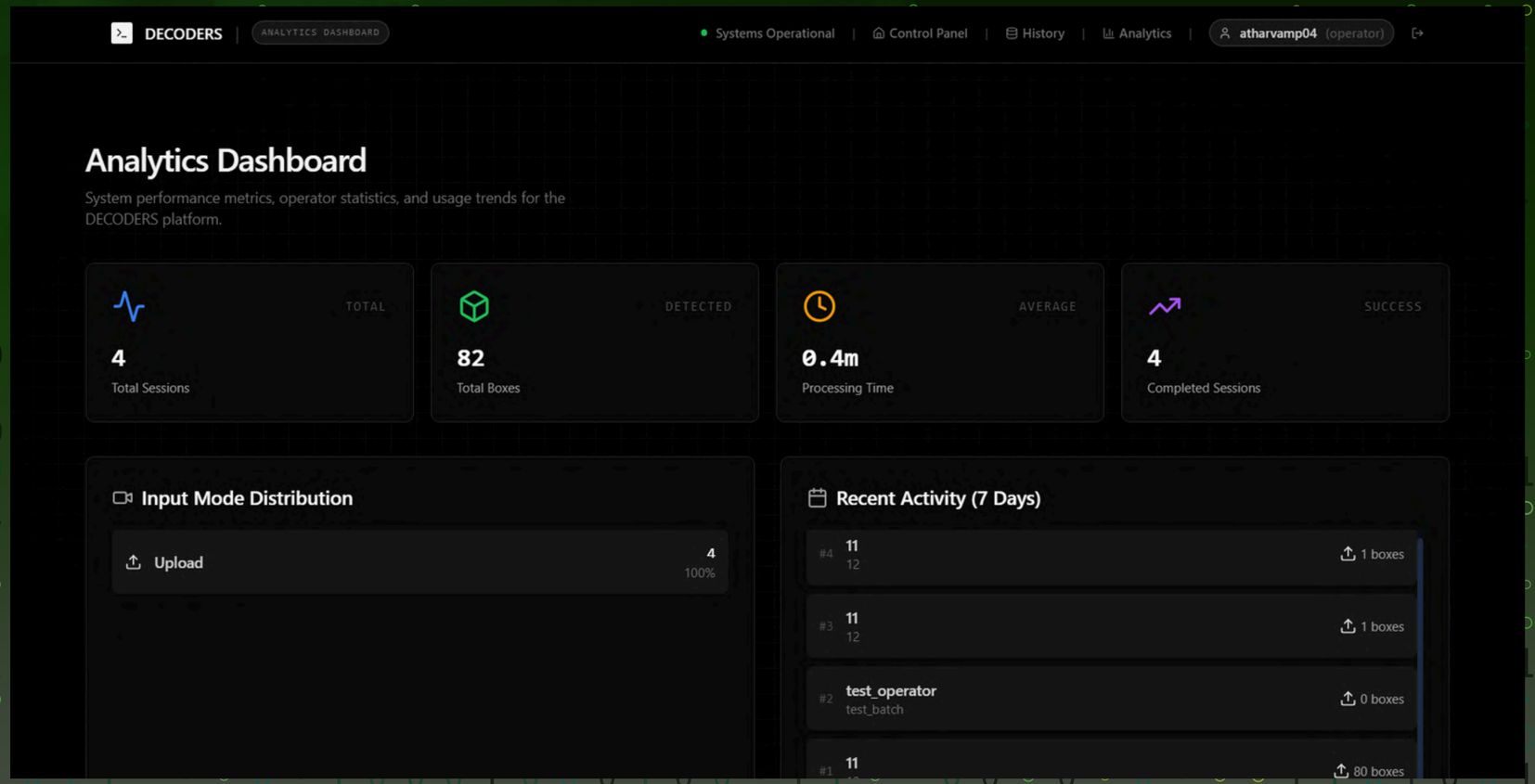
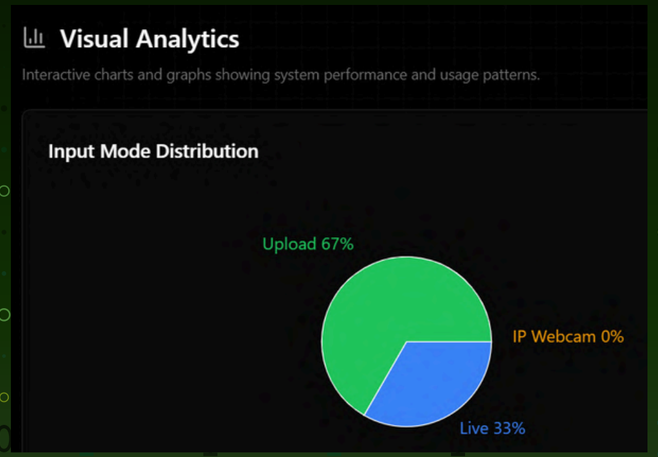
Analytics Dashboard





Real-time system metrics — total sessions, total boxes detected, average processing time, operator performance, and input mode distribution.